

② Technique 2: top-k sampling

↳ Restrict the sampled tokens to the top k most likely tokens, and exclude all other tokens

words "close" "every" "after" "found" "index" "house" "pizza" "house" "you"
indices 0 1 2 3 4 5 6 7 8

Logits = [4.51, 0.89, -1.90, 6.75, 1.63, -1.62, 1.89, 6.28, 1.79]

top-k=3 → [4.51] [6.75] [6.28]

→ Inf mask [4.51, -inf, -inf, 6.75, -inf, -inf, -inf, 6.28, -inf]

Softmax [0.00, 0, 0, 0.57, 0, 0, 0, 0.36, 0]

By assigning zero probabilities to the non-top-k positions, we ensure that next token is always sampled from a top-k position

→ Logits

Temperature
↓
Sample from multinomial

top-k sampling
+ temperature scaling

⇒ Modified text generation function

Logits → Top-k Sampling → $\frac{\text{Logits}}{\text{Temp}}$ → Softmax → Sample from multinomial distribution

Overfitting is reduced!

↳ True Generative model